

FEDERATED LEARNING OVER WEATHER STATION NETWORKS: PERSONALIZED TEMPERATURE FORECASTING FOR ENERGY DEMAND ESTIMATION

Madalina Harii

ABSTRACT

This report presents a federated learning (FL) system for next-day mean temperature forecasting across a network of nine Finnish Meteorological Institute (FMI) weather stations spanning coastal, inland, and sub-arctic climate regimes. Each station trains a personalized linear regression model on its local daily measurements, collaborating with geographically nearby stations via a weighted FL network without sharing raw data. Temperature predictions are translated into heating degree days (HDD) following the Finnish Meteorological Institute standard, providing a proxy for regional residential heating energy demand. We formulate the problem as Generalized Total Variation Minimization (GTVMIn) and solve it via a closed-form Ridge regression update applied synchronously across all nodes. Two FL systems are compared, one using a purely distance-based collaboration graph and one using a temperature-correlation-weighted graph, evaluated across three experiments varying training set size. FL improves over the local-only baseline only when training data is minimal (≈ 181 days), achieving a 2.1% reduction in mean test RMSE, while providing no benefit with one or two years of local data.

Index Terms— Federated learning, networks, personalized machine learning, trustworthy artificial intelligence

1. INTRODUCTION

Accurate short-term temperature forecasting is a critical input for regional energy demand planning. Utilities and grid operators rely on next-day temperature predictions to estimate residential heating requirements and balance load across regions. In Finland, heating demand varies substantially across the country: coastal areas benefit from the moderating effect of the Baltic Sea and its bays, while inland and northern regions experience lower and more volatile temperatures. A single global forecasting model trained on all stations simultaneously would fail to capture these local characteristics. Conversely, purely local models trained on a single station's historical record may be unreliable, particularly for stations with sparse or noisy measurements.

Federated Learning (FL) offers a solution to this tension [1]. In an FL system, each device - here, each weather station - trains its own personalized model on local data while

collaborating with neighboring devices by exchanging model parameters rather than raw measurements. This makes FL both privacy-friendly and decentralized: no raw meteorological data needs to leave each station. FL has been proposed for smart grid applications, where heterogeneous local conditions and data privacy requirements make centralized approaches impractical [1]. This project implements an FL system over a network of nine Finnish Meteorological Institute (FMI) weather stations to predict next-day mean temperature at each station. Predictions are translated into heating degree days (HDD) following the Finnish Meteorological Institute standard, providing a proxy for regional residential heating energy demand. The FL system is formulated as Generalized Total Variation Minimization (GTVMIn) [1], which balances local model fit against inter-station agreement, with collaboration strength proportional to geographic proximity. The remainder of this report is structured as follows. Section 2 formulates the FL problem, including network design, feature and label definitions, local models, and loss functions. Section 3 describes the GTVMIn-based methodology and the Ridge regression algorithm used to solve GTVMIn. Section 4 presents numerical experiments comparing two FL system designs across three data regimes. Section 5 concludes with a brief discussion of results and directions for future work.

2. PROBLEM FORMULATION

Raw data. Daily meteorological observations are downloaded from the Finnish Meteorological Institute (FMI) Open Data portal [3] for the period January 2022 to December 2024, covering three full years (1096 days). Hourly records are aggregated to daily values: daily mean temperature (t_{2m}), daily minimum temperature (t_{min}), and daily maximum temperature (t_{max}). Daily mean wind speed (ws_{10min}) was downloaded but excluded from the feature set because two stations (Tampere Härmälä and Inari Saariselkä) have no anemometer, and all FL nodes must share identical feature dimensionality for the GTVMIn weight vectors to be comparable across nodes.

Station selection. Nine FMI stations are selected representing three climatically distinct regimes across Finland:

- **Coastal:** Helsinki Kaisaniemi (fmid: 100971), Turku Artukainen (100949), Oulu Vihreäsaari (101794)

- **Inland:** Tampere Härmälä (101124), Jyväskylä Airport (137208), Kuopio Maaninka (101572)
- **Northern:** Rovaniemi Apukka (101933), Sodankylä (101932), Inari Saariselkä (102005)

This selection spans approximately 60°N to 68°N latitude, capturing maritime south-west, continental central, and sub-arctic climate regimes. Stations were selected to ensure geographic spread and avoid clustering, so that the FL network captures genuine climate heterogeneity rather than redundant nearby measurements.

Local data points. Each node i holds a local dataset $\mathcal{D}^{(i)} = \{(\mathbf{x}^{(i,t)}, y^{(i,t)})\}_{t=1}^{m_i}$, where the feature vector $\mathbf{x}^{(i,t)} \in \mathbb{R}^3$ for day t is:

$$\mathbf{x}^{(i,t)} = [T_{\text{mean}}^{(t)}, T_{\text{min}}^{(t)}, T_{\text{max}}^{(t)}]^\top \quad (1)$$

and the label $y^{(i,t)} = T_{\text{mean}}^{(t+1)}$ is the next-day mean temperature. Today's mean, minimum, and maximum temperature are used as features due to their direct physical relationship to next-day temperature: the daily temperature range captures atmospheric stability, and persistence of temperature is the strongest single predictor of next-day conditions. Predicting temperature directly is preferred over predicting HDD directly because temperature is a clean, continuously measured variable with well-understood statistical properties, making it more suitable as a regression target.

Heating degree days are recovered from predictions as a post-processing step, following the Finnish Meteorological Institute standard [4]:

$$\text{HDD} = \begin{cases} \max(17 - \hat{y}, 0) & \text{if } \hat{y} < 12^\circ\text{C} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $\hat{y}$ is the predicted next-day mean temperature. This standard uses an indoor baseline of 17°C and a heating activation threshold of 12°C, reflecting that Finnish buildings do not require active heating above this threshold due to internal heat gains from occupants and appliances [4]. HDD results on the test set are reported in Section 4.

FL network. Each node $i \in \mathcal{V}$ corresponds to one FMI station. Edges $\{i, i'\} \in \mathcal{E}$ are defined between all pairs of stations within a geographic distance threshold of $d_{\text{max}} = 300$ km, computed using the Haversine formula from station coordinates. Edge weights are set as:

$$A_{i,i'} = \exp\left(-\frac{d(i,i')^2}{2\sigma^2}\right), \quad \sigma = 150 \text{ km} \quad (3)$$

where $d(i,i')$ is the great-circle distance between stations i and i' . This Gaussian kernel assigns higher collaboration weight to nearby stations and decays smoothly with distance, reflecting the spatial autocorrelation of temperature fields. The values $\sigma = 150$ km and $d_{\text{max}} = 300$ km are fixed design

choices: d_{max} ensures each station has at least two neighbours given the selected station layout, and σ is set to half the distance threshold so that boundary neighbours receive a weight of $\exp(-2) \approx 0.14$. Stations beyond 300 km receive no edge. This distance-based graph defines **System A** and serves as the baseline for comparison in Section 4.

Local models. Each node i trains a linear regression model with parameter vector $\mathbf{w}^{(i)} \in \mathbb{R}^3$, producing predictions $\hat{y}^{(i)} = \mathbf{x}^\top \mathbf{w}^{(i)}$. Features are standardised to zero mean and unit variance using statistics computed on the training set only, making an explicit bias term redundant. Linear models are chosen for two reasons: first, they are interpretable - the learned weights directly reflect how each station weighs today's temperatures to predict tomorrow's; second, daily mean temperature is well-approximated as a linear combination of same-day measurements [2].

Local loss functions. Each node minimises the mean squared error (MSE) on its local dataset:

$$L^{(i)}(\mathbf{w}^{(i)}) = \frac{1}{m_i} \|\mathbf{y}^{(i)} - \mathbf{X}^{(i)} \mathbf{w}^{(i)}\|_2^2 \quad (4)$$

MSE is chosen over mean absolute error (MAE) for two reasons. First, MSE penalises large prediction errors more heavily, which is desirable for energy planning where extreme cold events carry disproportionate consequences for heating demand. Second, MSE yields a smooth and strongly convex local loss function, which is a required property for the convergence guarantees of the Ridge-based FL algorithm described in Section 3 [1].

3. METHODOLOGY

Variation measure. We formulate the FL problem as Generalized Total Variation Minimization (GTVMIn) [1], which balances local model fit at each station against agreement between neighbouring stations. The variation between neighbouring nodes i and i' is measured using the squared Euclidean distance between their local model parameters:

$$d(\mathbf{w}^{(i)}, \mathbf{w}^{(i')}) = \|\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\|_2^2 \quad (5)$$

This choice is motivated by two properties. First, it is a natural and symmetric measure of disagreement for real-valued parameter vectors. Second, combined with the MSE local loss functions defined in Section 2, it renders the full GTVMIn objective smooth and strongly convex, enabling a closed-form solution at each node [1].

The full GTVMIn objective over all $n = 9$ nodes is:

$$\min_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(n)}} \underbrace{\sum_{i=1}^n L^{(i)}(\mathbf{w}^{(i)})}_{\text{local fit}} + \alpha \underbrace{\sum_{\{i,i'\} \in \mathcal{E}} A_{i,i'} \|\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\|_2^2}_{\text{GTV penalty}} \quad (6)$$

where $\alpha > 0$ controls the trade-off between local fit and inter-station agreement. When $\alpha = 0$, each station trains independently; as $\alpha \rightarrow \infty$, all stations collapse toward a single shared model. The value of α is treated as a hyperparameter selected by minimising validation loss independently for each FL system.

FL algorithm - Ridge reformulation. Rather than solving GTVMin via gradient descent, we exploit the fact that the GTVMin objective is separable: fixing all nodes except i , the subproblem for node i is:

$$\min_{\mathbf{w}^{(i)}} \left\| \mathbf{X}^{(i)} \mathbf{w}^{(i)} - \mathbf{y}^{(i)} \right\|_2^2 + \alpha s_i \left\| \mathbf{w}^{(i)} - \bar{\boldsymbol{\theta}}_i \right\|_2^2 \quad (7)$$

where

$$s_i = \sum_{i' \in \mathcal{N}(i)} A_{i,i'}, \quad \bar{\boldsymbol{\theta}}_i = \frac{1}{s_i} \sum_{i' \in \mathcal{N}(i)} A_{i,i'} \mathbf{w}^{(i')} \quad (8)$$

are the neighbourhood strength and the weighted average of neighbour parameters respectively. Substituting $\mathbf{u} = \mathbf{w}^{(i)} - \bar{\boldsymbol{\theta}}_i$ yields a standard Ridge regression problem:

$$\min_{\mathbf{u}} \left\| \mathbf{X}^{(i)} \mathbf{u} - \tilde{\mathbf{y}}^{(i)} \right\|_2^2 + \alpha s_i \left\| \mathbf{u} \right\|_2^2, \quad \tilde{\mathbf{y}}^{(i)} = \mathbf{y}^{(i)} - \mathbf{X}^{(i)} \bar{\boldsymbol{\theta}}_i \quad (9)$$

This is solved in closed form via `scikit-learn`:

$$\text{Ridge}(\alpha s_i).fit(\mathbf{X}^{(i)}, \tilde{\mathbf{y}}^{(i)})$$

The recovered parameters are $\mathbf{w}^{(i)} = \mathbf{u}^* + \bar{\boldsymbol{\theta}}_i$. This reformulation requires no learning rate and converges to the global GTVMin optimum in a finite number of rounds [1].

The algorithm is implemented as synchronous message passing in three steps per round [1]:

1. **Share:** every node i broadcasts its current parameters $\mathbf{w}^{(i,t)}$ to all neighbours $i' \in \mathcal{N}(i)$.
2. **Update:** every node i computes s_i , $\bar{\boldsymbol{\theta}}_i$, and the modified labels $\tilde{\mathbf{y}}^{(i)}$, then solves the Ridge subproblem to obtain $\mathbf{w}^{(i,t+1)}$.
3. **Repeat** for $T = 50$ rounds.

All nodes are updated simultaneously using parameters from round t only - not mid-round updates. No raw meteorological data is exchanged at any step - only parameter vectors $\mathbf{w}^{(i)} \in \mathbb{R}^3$. All nodes are initialised at $\mathbf{w}^{(i,0)} = \mathbf{0}$.

Two FL systems compared. To evaluate the impact of graph construction on forecasting performance, we compare two FL systems differing only in edge weight definition, while keeping local models, loss functions, and algorithm identical:

- **System A - Distance graph** (baseline): edge weights defined by geographic proximity via the Gaussian kernel described in Section 2. This graph encodes the assumption that spatially close stations share similar temperature dynamics.

- **System B - Correlation graph:** edge weights are re-scaled by the Pearson correlation of daily mean temperatures between station pairs, computed on the training set only to avoid data leakage:

$$A_{i,i'}^{(B)} = \max\left(0, \rho\left(T^{(i)}, T^{(i')}\right)\right) \cdot \exp\left(-\frac{d(i,i')^2}{2\sigma^2}\right) \quad (10)$$

Pairs with zero or negative correlation receive no edge. System B encodes the assumption that stations with empirically similar temperature dynamics should collaborate more strongly, regardless of distance alone.

Both systems share the same 15 edges - all station pairs within 300 km exhibit positive Pearson correlation on the training data, reflecting the shared seasonal temperature cycle across Finland. System B edge weights are slightly lower on average (0.15–0.81 vs. 0.15–0.82 for System A) because the correlation multiplier scales the Gaussian weights down marginally. By comparing Systems A and B we isolate the effect of correlation-based re-weighting on FL performance [1].

4. NUMERICAL EXPERIMENTS

Experimental setup. Three experiments are conducted varying the size of the local training sets to investigate under which conditions GTVMin-based FL provides benefit over local-only Ridge regression. Data are split chronologically to respect the temporal structure of the time series, with no shuffling [2]:

- **Exp. 1 - Full data:** train 2022-01-01 to 2023-12-31 (≈ 730 days), val 2024-01-01 to 2024-06-30, test 2024-07-01 to 2024-12-31.
- **Exp. 2 - Reduced data:** train 2022-01-01 to 2022-12-31 (≈ 365 days), val 2023-01-01 to 2023-06-30, test 2023-07-01 to 2023-12-31.
- **Exp. 3 - Minimal data:** train 2022-01-01 to 2022-06-30 (≈ 181 days), val 2022-07-01 to 2022-09-30, test 2022-10-01 to 2022-12-31.

Each experiment evaluates three systems: a **Baseline** (local-only Ridge, $\alpha = 0$, no collaboration), **System A** (GTVMin, distance-based graph), and **System B** (GTVMin, correlation-weighted graph). Features are standardised using training-set statistics only; labels remain in $^{\circ}\text{C}$ so that RMSE is directly interpretable as a temperature prediction error. The regularisation parameter α is selected independently per system by minimising mean validation MSE over the grid $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 0.05, 0.1, 0.5, 1, 5, 10, 50, 100, 500\}$, with $T = 50$ synchronous rounds. Each station's data is split and normalised independently, consistent with the federated setting where no raw data is shared across nodes.

Results. Table 1 reports mean train, validation, and test RMSE across all nine stations for each experiment and system. Table 2 reports mean daily HDD on the test set (Exp. 1) computed from model predictions following the FMI standard [4].

Table 1: Mean RMSE ($^{\circ}\text{C}$) across nine stations. α^* selected on validation set independently per system.

Exp.	System	Train	Val	Test	α^*
1	Baseline	5.24	5.58	5.29	-
	System A	5.24	5.58	5.30	50
	System B	5.24	5.58	5.30	50
2	Baseline	5.39	5.34	5.47	-
	System A	5.39	5.34	5.47	10^{-4}
	System B	5.39	5.34	5.47	10^{-4}
3	Baseline	4.29	3.58	4.15	-
	System A	4.34	3.46	4.07	500
	System B	4.34	3.47	4.07	500

Table 2: Mean daily HDD ($^{\circ}\text{C}\cdot\text{days}$) and MAE on the test set (Exp. 1), System A. Baseline and System B differ by $< 0.1^{\circ}\text{C}\cdot\text{days}$ at all stations.

Station	Actual	Predicted	MAE
Helsinki Kaisaniemi	6.24	12.72	6.48
Turku Artukainen	6.52	13.03	6.51
Oulu Vihreäsaari	8.43	11.59	3.29
Tampere Härmälä	7.68	13.01	5.45
Jyväskylä Airport	8.74	12.85	4.31
Kuopio Maaninka	8.61	12.28	3.81
Rovaniemi Apukka	11.12	12.69	2.85
Sodankylä	11.49	12.60	2.87
Inari Saariselkä	12.33	12.37	2.24

Discussion. In Exp. 1 and 2, FL provides no measurable benefit over the Baseline. In Exp. 2, the algorithm selects $\alpha^* = 10^{-4}$, effectively choosing no collaboration, confirming that 365 training days are sufficient for a well-determined 3-feature linear model. In Exp. 1, $\alpha^* = 50$ is selected but still fails to improve test RMSE, suggesting that strong regularisation forces parameters toward a consensus that is suboptimal when local data is plentiful.

Exp. 3 reveals the regime where FL is beneficial. System A achieves a test RMSE of 4.07°C versus the Baseline’s 4.15°C (2.1% improvement) with only 181 training days. The large optimal $\alpha^* = 500$ indicates that strong neighbour regularisation compensates for sparse local data. The largest per-station gains occur at inland stations (Tampere: $4.92 \rightarrow 4.69^{\circ}\text{C}$, Kuopio: $3.70 \rightarrow 3.31^{\circ}\text{C}$, Jyväskylä: $3.99 \rightarrow 3.61^{\circ}\text{C}$), which benefit from sharing signal with climatically similar neighbours. Helsinki and Turku are the only stations where FL slightly increases test RMSE, reflecting their distinct coastal climate relative to inland neighbours.

System A and System B perform nearly identically across all experiments (difference $< 0.002^{\circ}\text{C}$ RMSE). Since all station pairs within 300 km exhibit positive Pearson correlation, System B retains the same 15 edges as System A with marginally lower weights. The negligible performance gap indicates that geographic proximity is a sufficient proxy for climate similarity in this network.

Heating degree days. Table 2 reports mean daily HDD computed from model predictions following the FMI standard [4]. All three systems systematically over-estimate HDD at southern stations (Helsinki MAE: $6.5^{\circ}\text{C}\cdot\text{days}$, Turku: $6.5^{\circ}\text{C}\cdot\text{days}$), while northern stations are well calibrated (Inari MAE: $2.2^{\circ}\text{C}\cdot\text{days}$). This north-south gradient reflects the test period (July-December 2024): southern stations have many summer days where actual HDD = 0, but the linear model - trained mainly on colder months - under-predicts warm temperatures and over-estimates heating demand. Northern stations experience fewer warm days in the test period, reducing this bias. All three systems produce near-identical HDD estimates, consistent with their similar temperature RMSE.

5. CONCLUSION

This report presented a GTVMin-based federated learning system for next-day temperature forecasting across nine Finnish Meteorological Institute weather stations, with predictions translated into heating degree days following the FMI standard as a proxy for regional residential heating energy demand. Two FL systems were compared - System A using a distance-based collaboration graph and System B using a correlation-weighted graph - evaluated across three experiments varying training set size.

The central finding is that FL benefit depends strongly on local data availability. With 365 or more training days, a 3-feature linear model is well-determined locally and neighbour collaboration provides no improvement - the algorithm itself selects near-zero regularisation ($\alpha^* = 10^{-4}$) in the reduced-data setting, effectively choosing independence. With only 181 training days, System A achieves a 2.1% reduction in mean test RMSE over the local-only baseline (4.07°C vs. 4.15°C), with the largest gains at inland stations that share similar continental climate with their neighbours. The two FL systems perform nearly identically across all experiments, indicating that geographic proximity is a sufficient proxy for climate similarity in this network and correlation-based re-weighting provides no additional benefit [1].

The HDD analysis reveals a systematic over-estimation of heating demand at southern stations (MAE up to $6.5^{\circ}\text{C}\cdot\text{days}$), driven by the linear model’s inability to predict summer temperatures correctly when trained predominantly on cold-season data. Northern stations are well calibrated (Inari MAE: $2.2^{\circ}\text{C}\cdot\text{days}$), as they experience fewer warm days in the test period.

Several limitations point to directions for future work.

First, the 3-feature linear model is a strong simplification; augmenting features with lagged temperature values or polynomial interactions would increase model complexity and expand the data regime where FL collaboration is beneficial [2]. Second, wind speed data was unavailable for two stations, reducing feature informativeness; a more complete sensor network would provide richer local signals. Third, the comparison between Systems A and B was limited by the fact that all station pairs within 300 km have positive temperature correlation, meaning both graphs share identical edge structure; a stricter correlation threshold would produce genuinely different topologies and a more informative comparison. Finally, the seasonal HDD bias suggests that separate models for heating and non-heating seasons, or a non-linear hypothesis class [2], would better capture the summer-winter transition and improve energy demand estimates.

6. REFERENCES

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